

Bit manipulation

*set **i(th)** Bit*

you have given a number and position of the bit.

Examples: **N=5, i=1**

i=1

• **N=5** $\longrightarrow$ **1 0 1**

- set the bit means, 0 is converted into 1 OR 1 is converted into 0.

1 0 1 $\longrightarrow$ **N=5**

101

N=5

OR

010

(1<<i)

111

result (7)

- Write (1<<i).
- do, N | (1<<i)



```
1  // C++ program to set K-th bit of a number N
2
3  #include <iostream>
4  using namespace std;
5
6  void setIthBit(int &n, int i)
7  {
8      int mask = (1 << i);
9      n = (n | mask);
10 }
11
12 int main()
13 {
14     int n, i;
15     cin >> n >> i;
16     setIthBit(n, i);
17     cout << n << endl;
18     return 0;
19 }
20
```

code in c++



```
1  function setIthBit(n, i) {  
2      let mask = (1 << i);  
3      return (n | mask);  
4  }  
5  
6  let n = parseInt(prompt("Enter the number:"));  
7  let i = parseInt(prompt("Enter the bit position:"));  
8  
9  n = setIthBit(n, i);  
10 console.log("Result:", n);
```

code in javascript



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